Trigger dev environment setup

1.1 Enabling TR debug tools

For this step to be possible we need to first understand hotkeys + configs and have the related files set up. Read and follow “*BANG documentation\Config documentation\Config basics”* and “*BANG documentation\Hotkey documentation\Hotkey basics”.*

You should now have a *user.con* and *user.cfg* and understand how to use them.

The are three **essential** configs we need to enable: *debugTriggers*, *generateTRConstants*, and *enableTriggerEcho*. These configs will enable all actual debug tools for us. There are more TR related configs, which can be found lower on this page, but they’re situational.

We **need** to create 1 custom hotkey because it is not bound by default:
map("*somekey*", "root", "toggleTREchoLog", true)

To verify our configs were correctly enabled and our hotkeys work we will run through all the debug tools while in-game:

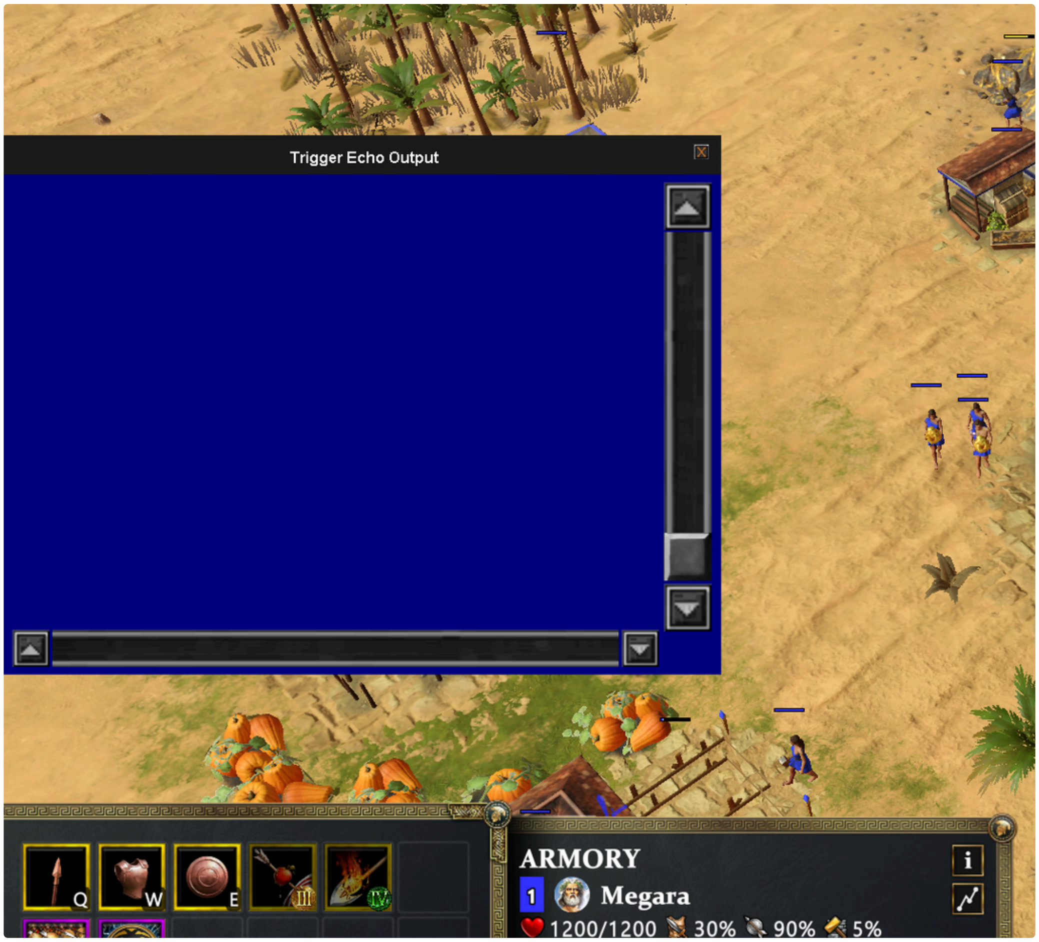
1. Restart your game to have the configs take effect.
2. Start any skirmish game, we don’t need a custom trigger file attached for this.

1.2 TR output menu

Verify we have this menu available to us.

Press the hotkey you chose during the first part of this setup

You should now see the menu come up, if it doesn’t either your *debugTriggers* config wasn’t properly set or the hotkey is conflicting/improperly set up.



In this menu all your trEcho will be displayed as long as you have the *enableTriggerEcho* config defined. It will also display all warnings generated by the engine.

1.3 TR constants

Verify we have this file available to us.

Start any game.

Navigate to: "*..\Users\<your username>\Games\Age of Mythology Retold\temp\Logs*"
Name: MythTRConstants.txt

If you don't see the text file present your *generateTRConstants* config wasn't properly set.

Many TR/KB syscalls want an ID as input instead of a name. To lookup what IDs map to what entity in the game we have the TR constants file. These constants act like what in coding is known as an "enum". Example usage:

kbUnitCount(cUnitTypeHillFort, cMyID, cUnitStateAlive)

In the example you can clearly read that we're trying to count Hill Forts. But cUnitTypeHillFort actually stands for 347 and that is also what you get if you echo that constant. We turn that number into a name because otherwise we have no idea what we're reading as humans. For example:

kbUnitCount(87, cMyID, 4)

The line above is counting Minotaurs that are dead, but how would you know that at a glance? This is why using the constants over actual numbers is always desirable. Maybe you also already noticed I changed cUnitStateAlive to a number, because that variable in fact is also a constant/enum.

End of in-game setup part, we have visited all trigger related tools and verified they work.

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2. Shared XS setup

Read the “*BANG documentation\XS documentation\How-to Doxygen”* document to get a better understanding of all possible syscalls.

Follow the steps in: “*BANG documentation\XS documentation\VS Code Extension For XS\Installing VS Code and extension for XS”*.

Don’t forget to also follow the steps in: “*BANG documentation\XS documentation\VS Code Extension For XS\VS Code adjust settings”*.

Read up on the XS Debugger in: “*BANG documentation\XS documentation\XS debugger”*.

3.1 TR configs

This is a table of all TR related configs:

Config	**Help text**
debugTriggers	Enable debugging of triggers.
patchTriggers	Scenario triggers get patched up based on the trigger XML file. For example you have a trigger but you change in the trigger_data.xml what it actually does. If you want your scenarios to be patched so that their existing triggers now actually perform the new logic you must enable this and resave your scenario.
generateTRConstants	Generates a text file including all external trigger constants.
enableTriggerEcho	Enables trigger-echo outputs.
OutputSlowRuleWarnings	In the XS debugger’s output window you will be notified if a rule took longer than 5ms to complete, which would mean it’s lagging the game more than we want.

3.2 Iteration speedup configs

This is a table of all iteration speedup related configs (not TR specific):

noIntroCinematics	Skips all videos displayed during game startup
disableAssetPreloading	Disables asset preloading when starting a game. Since you will be potentially reloading a map many times you don’t want to wait on this, you can use this for shorter load times.
runAsFastAsPossible	Tries to run the game faster than normal, to test AI logic quicker. How much faster the game runs is capped by your CPU speed.

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4. Important TR hotkeys

If you wish to not use the default binds then copy the lines in the left column to the *user.con* and change “*somekey*” to your wanted binding. You can look in *game.con* for examples of possible bindings.

map("*somekey*", "root", "toggleTREchoLog", true)	Opens the debug window associated with the *enableTriggerEcho*config. **Unbound by default.**
map("somekey", "root", "gadgetToggle("XSDebugger")")	Open the debug window associated with debugging XS, **alt-shift-d is the default binding**.
map ("somekey", "world", "player(1)")	Switches to player 1 in SP games, **control-f1 is the default binding scheme, control-f2 for player 2 etc.** Switching to Mother nature has a different default: map ("control-shift-f10", "world", "player(0)")